EXPT:- 05
06-08-2026.

EXPERIMENT ON PACKET TOOL.

Aim:-
To understand the feature of wireshark. as a packet capture tool and understand encapsulation of information at various layers of a protocol stack.

Packet Sniffer:-

* Capture packets sent and recieved by the compute.
* Displays protocols information of captured packets
* Works passively.
* Recieves copies of nehoorks packets for analysis.

Packet Sniffer Tools:-

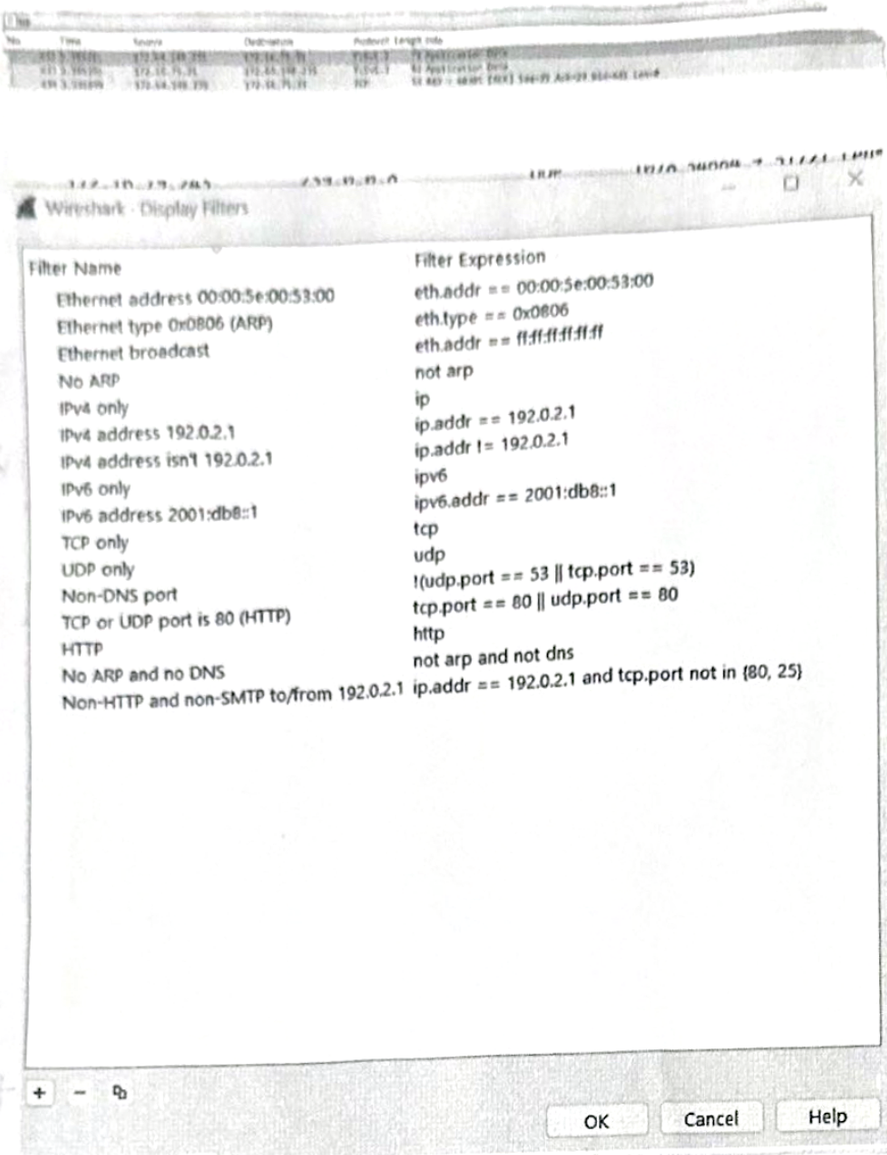
* Tcpdump :- Command line packet capture tool.
* Wireshark :- Graphical packet analyser.

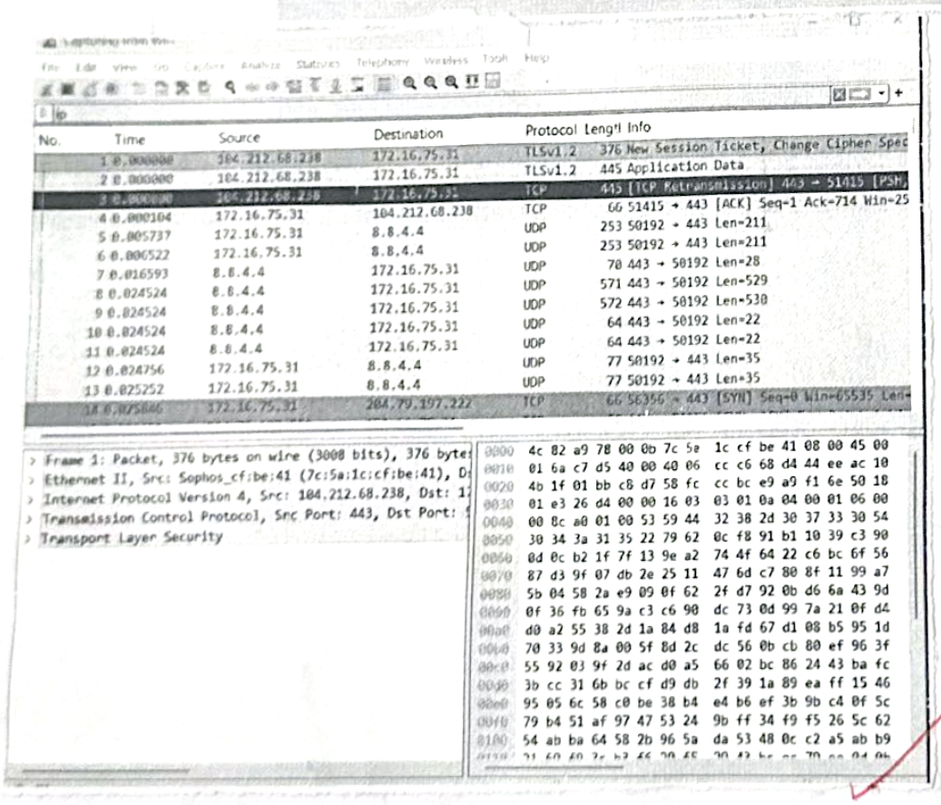
Description:-

Wireshark is a network protocols analyser that captures and displays nehoorks packets in real time It helps monitor, analyze, troubleshoot, and inspect network traffic using filters and packet details.

Uses of nehoorks:

* Capture nehoorks traffic.
* Analyzes protocols.
* Applies displays filters.
* monitors nehoorks issues.


Wireshark · Display Filters
Filter Name
Filter Expression
Ethernet address 00:00:5e:00:53:00
eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)
eth.type == 0x0806
Ethernet broadcast
eth.addr == ff:ff:ff:ff:ff:ff
No ARP
not arp
IPv4 only
ip
IPv4 address 192.0.2.1
ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1
ip.addr != 192.0.2.1
IPv6 only
ipv6
IPv6 address 2001:db8::1
ipv6.addr == 2001:db8::1
TCP only
tcp
UDP only
udp
Non-DNS port
!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)
tcp.port == 80 || udp.port == 80
HTTP
http
No ARP and no DNS
not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1
ip.addr == 192.0.2.1 and tcp.port not in {80, 25}
OK
Cancel
Help


No. Time Source Destination Protocol Length Info
1 0.000000 104.212.68.238 172.16.75.31 TLSv1.2 376 New Session Ticket, Change Cipher Spec
2 0.000000 104.212.68.238 172.16.75.31 TLSv1.2 445 Application Data
3 0.000000 104.212.68.238 172.16.75.31 TCP 445 [TCP Retransmission] 443 → 51415 [PSH,
4 0.000104 172.16.75.31 104.212.68.238 TCP 66 51415 → 443 [ACK] Seq=1 Ack=714 Win=25
5 0.005737 172.16.75.31 8.8.4.4 UDP 253 50192 → 443 Len=211
6 0.006522 172.16.75.31 8.8.4.4 UDP 253 50192 → 443 Len=211
7 0.016593 8.8.4.4 172.16.75.31 UDP 70 443 → 50192 Len=28
8 0.024524 8.8.4.4 172.16.75.31 UDP 571 443 → 50192 Len=529
9 0.024524 8.8.4.4 172.16.75.31 UDP 572 443 → 50192 Len=530
10 0.024524 8.8.4.4 172.16.75.31 UDP 64 443 → 50192 Len=22
11 0.024524 8.8.4.4 172.16.75.31 UDP 64 443 → 50192 Len=22
12 0.024756 172.16.75.31 8.8.4.4 UDP 77 50192 → 443 Len=35
13 0.025252 172.16.75.31 8.8.4.4 UDP 77 50192 → 443 Len=35
14 0.025846 172.16.75.31 204.79.197.222 TCP 66 56356 → 443 [SYN] Seq=0 Win=65535 Len=
Frame 1: Packet, 376 bytes on wire (3008 bits), 376 byte
Ethernet II, Src: Sophos_cf:be:41 (7c:5a:1c:cf:be:41), D
Internet Protocol Version 4, Src: 104.212.68.238, Dst: 1
Transmission Control Protocol, Src Port: 443, Dst Port:
Transport Layer Security
0000 4c 82 a9 78 00 0b 7c 5a 1c cf be 41 08 00 45 00
0010 01 6a c7 d5 40 00 40 06 cc c6 68 d4 44 ee ac 10
0020 4b 1f 01 bb c8 d7 58 fc cc bc e9 a9 f1 6e 50 18
0030 01 e3 26 d4 00 00 16 03 03 01 0a 04 00 01 06 00
0040 00 8c a0 01 00 53 59 44 32 38 2d 30 37 33 30 54
0050 30 34 3a 31 35 22 79 62 0c f8 91 b1 10 39 c3 90
0060 0d 0c b2 1f 7f 13 9e a2 74 4f 64 22 c6 bc 6f 56
0070 87 d3 9f 07 db 2e 25 11 47 6d c7 80 8f 11 99 a7
0080 5b 04 58 2a e9 09 0f 62 2f d7 92 0b d6 6a 43 9d
0090 0f 36 fb 65 9a c3 c6 90 dc 73 0d 99 7a 21 0f d4
00a0 d0 a2 55 38 2d 1a 84 d8 1a fd 67 d1 08 b5 95 1d
00b0 70 33 9d 8a 00 5f 8d 2c dc 56 0b cb 80 ef 96 3f
00c0 55 92 03 9f 2d ac d0 a5 66 02 bc 86 24 43 ba fc
00d0 3b cc 31 6b bc cf d9 db 2f 39 1a 89 ea ff 15 46
00e0 95 05 6c 58 c0 be 38 b4 e4 b6 ef 3b 9b c4 0f 5c
00f0 79 b4 51 af 97 47 53 24 9b ff 34 f9 f5 26 5c 62
0100 54 ab ba 64 58 2b 96 5a da 53 48 0c c2 a5 ab b9

Wireshark · Display Filters

Filter Name	Filter Expression
Ethernet address 00:00:5e:00:53:00	eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)	eth.type == 0x0806
Ethernet broadcast	eth.addr == ff:ff:ff:ff:ff:ff
No ARP	not arp
IPv4 only	ip
IPv4 address 192.0.2.1	ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1	ip.addr != 192.0.2.1
IPv6 only	ipv6
IPv6 address 2001:db8::1	ipv6.addr == 2001:db8::1
TCP only	tcp
UDP only	udp
Non-DNS port	!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)	tcp.port == 80 || udp.port == 80
HTTP	http
No ARP and no DNS	not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1	ip.addr == 192.0.2.1 and tcp.port not in {80, 25}

+ − ⧉ OK Cancel Help

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	104.212.68.238	172.16.75.31	TLSv1.2	376	New Session Ticket, Change Cipher Spec
2	0.000000	104.212.68.238	172.16.75.31	TLSv1.2	445	Application Data
3	0.000000	104.212.68.238	172.16.75.31	TCP	445	[TCP Retransmission] 443 → 51415 [PSH,
4	0.000104	172.16.75.31	104.212.68.238	TCP	66	51415 → 443 [ACK] Seq=1 Ack=714 Win=25
5	0.005737	172.16.75.31	8.8.4.4	UDP	253	50192 → 443 Len=211
6	0.006522	172.16.75.31	8.8.4.4	UDP	253	50192 → 443 Len=211
7	0.016593	8.8.4.4	172.16.75.31	UDP	78	443 → 50192 Len=28
8	0.024524	8.8.4.4	172.16.75.31	UDP	571	443 → 50192 Len=529
9	0.024524	8.8.4.4	172.16.75.31	UDP	572	443 → 50192 Len=530
10	0.024524	8.8.4.4	172.16.75.31	UDP	64	443 → 50192 Len=22
11	0.024524	8.8.4.4	172.16.75.31	UDP	64	443 → 50192 Len=22
12	0.024756	172.16.75.31	8.8.4.4	UDP	77	50192 → 443 Len=35
13	0.025252	172.16.75.31	8.8.4.4	UDP	77	50192 → 443 Len=35
14	0.025846	172.16.75.31	204.79.197.222	TCP	66	56356 → 443 [SYN] Seq=0 Win=65535 Len=

> Frame 1: Packet, 376 bytes on wire (3008 bits), 376 bytes
> Ethernet II, Src: Sophos_cf:be:41 (7c:5a:1c:cf:be:41), D
> Internet Protocol Version 4, Src: 104.212.68.238, Dst: 1
> Transmission Control Protocol, Src Port: 443, Dst Port:
> Transport Layer Security

```
0000  4c 82 a9 78 00 0b 7c 5a  1c cf be 41 08 00 45 00
0010  01 6a c7 d5 40 00 40 06  cc c6 68 d4 44 ee ac 10
0020  4b 1f 01 bb c8 d7 58 fc  cc bc e9 a9 f1 6e 50 18
0030  01 e3 26 d4 00 00 16 03  03 01 0a 04 00 01 06 00
0040  00 8c a0 01 00 53 59 44  32 38 2d 30 37 33 30 54
0050  30 34 3a 31 35 22 79 62  0c f8 91 b1 10 39 c3 90
0060  0d 0c b2 1f 7f 13 9e a2  74 4f 64 22 c6 bc 6f 56
0070  87 d3 9f 07 db 2e 25 11  47 6d c7 80 8f 11 99 a7
0080  5b 04 58 2a e9 09 0f 62  2f d7 92 0b d6 6a 43 9d
0090  0f 36 fb 65 9a c3 c6 90  dc 73 0d 99 7a 21 0f d4
00a0  d0 a2 55 38 2d 1a 84 d8  1a fd 67 d1 08 b5 95 1d
00b0  70 33 9d 8a 00 5f 8d 2c  dc 56 0b cb 80 ef 96 3f
00c0  55 92 03 9f 2d ac d0 a5  66 02 bc 86 24 43 ba fc
00d0  3b cc 31 6b bc cf d9 db  2f 39 1a 89 ea ff 15 46
00e0  95 05 6c 58 c0 be 38 b4  e4 b6 ef 3b 9b c4 0f 5c
00f0  79 b4 51 af 97 47 53 24  9b ff 34 f9 f5 26 5c 62
0100  54 ab ba 64 58 2b 96 5a  da 53 48 0c c2 a5 ab b9
```

Applications:-

* Network troubleshooting.
* Security analysis.
* Protocol debugging.
* Learning network protocols.

Capturing Packet:-

1. Open wireshark.
2. Select the network interface.
3. Click start capture.
4. Observe packet in real time.
5. Click Stop and save the capture.

Wireshark powers:-

1. Packet List:- Display captures packets.
2. Packet details:- shows protocols information.
3. Packet Bytes:- Display packet data in hexadecimal poin

Color Coding:

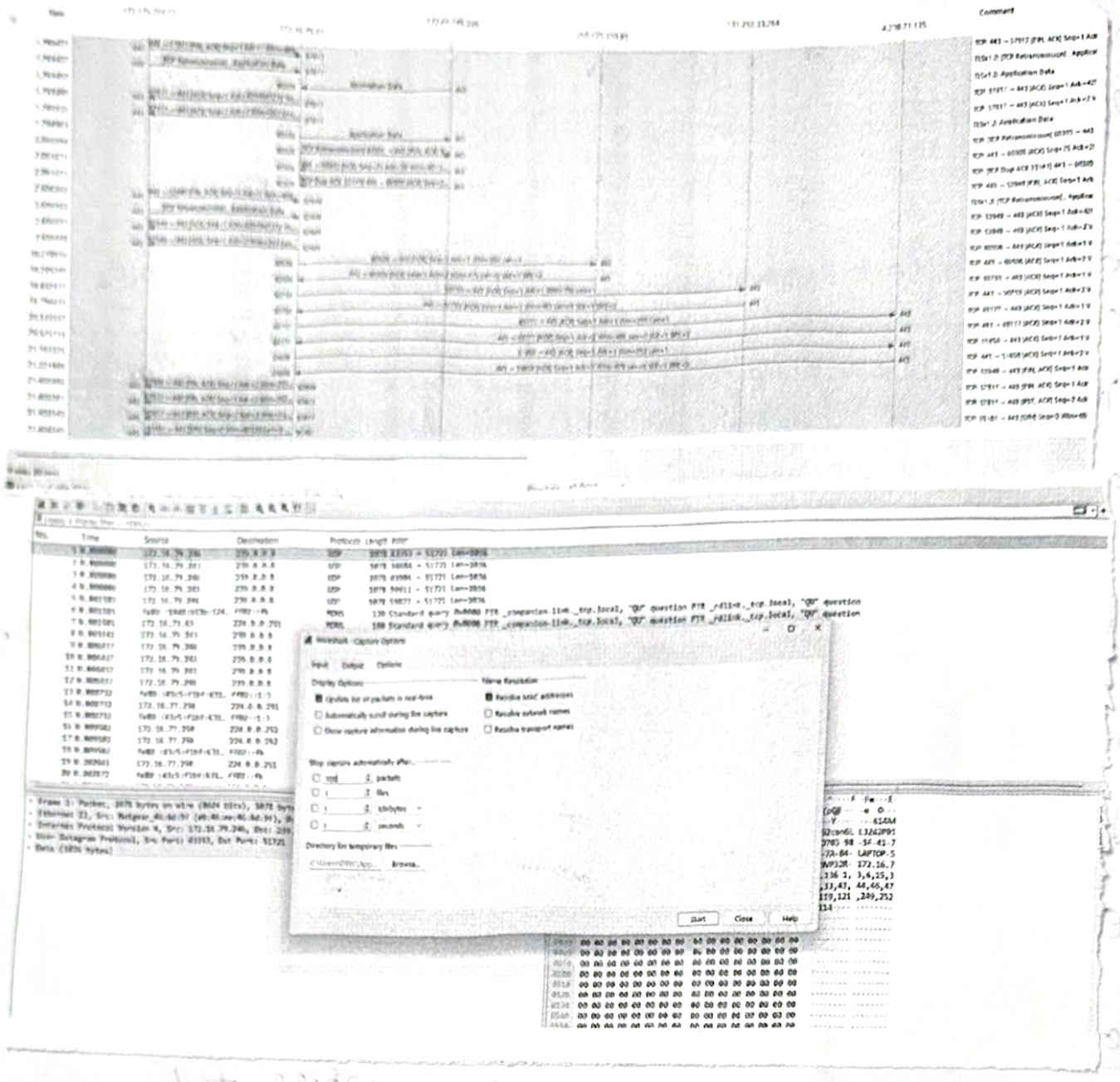
* Purple: TCP.
* Blue:- UDP.
* Black:- Error packets.

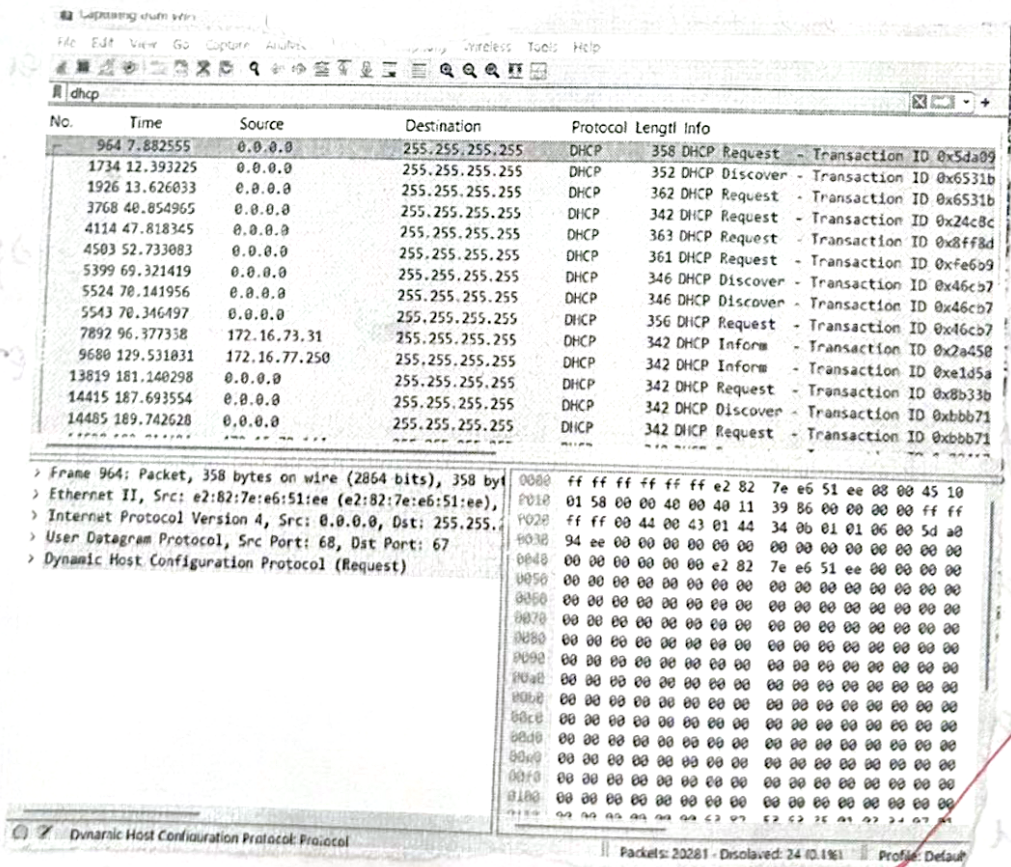
Filtering Packets:-

* Enter a filter:- in the filter bar.
* Right click to apply filters or follows the TCP stream.

Flow Graph:-

Displays the communication flow between devices making packing, exchange easier to understand.




dhcp
No. Time Source Destination Protocol Length Info
964 7.882555 0.0.0.0 255.255.255.255 DHCP 358 DHCP Request - Transaction ID 0x5da09
1734 12.393225 0.0.0.0 255.255.255.255 DHCP 352 DHCP Discover - Transaction ID 0x6531b
1926 13.626033 0.0.0.0 255.255.255.255 DHCP 362 DHCP Request - Transaction ID 0x6531b
3768 40.854965 0.0.0.0 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x24c8c
4114 47.818345 0.0.0.0 255.255.255.255 DHCP 363 DHCP Request - Transaction ID 0x8ff8d
4503 52.733083 0.0.0.0 255.255.255.255 DHCP 361 DHCP Request - Transaction ID 0xfe6b9
5399 69.321419 0.0.0.0 255.255.255.255 DHCP 346 DHCP Discover - Transaction ID 0x46cb7
5524 70.141956 0.0.0.0 255.255.255.255 DHCP 346 DHCP Discover - Transaction ID 0x46cb7
5543 70.346497 0.0.0.0 255.255.255.255 DHCP 356 DHCP Request - Transaction ID 0x46cb7
7892 96.377338 172.16.73.31 255.255.255.255 DHCP 342 DHCP Inform - Transaction ID 0x2a458
9680 129.531031 172.16.77.250 255.255.255.255 DHCP 342 DHCP Inform - Transaction ID 0xe1d5a
13819 181.140298 0.0.0.0 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x8b33b
14415 187.693554 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0xbbb71
14485 189.742628 0.0.0.0 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0xbbb71
Frame 964: Packet, 358 bytes on wire (2864 bits), 358 byt
Ethernet II, Src: e2:82:7e:e6:51:ee (e2:82:7e:e6:51:ee),
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Request)
0000 ff ff ff ff ff ff e2 82 7e e6 51 ee 08 00 45 10
0010 01 58 00 00 40 00 40 11 39 86 00 00 00 00 ff ff
0020 ff ff 00 44 00 43 01 44 34 0b 01 01 06 00 5d a0
0030 94 ee 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 e2 82 7e e6 51 ee 00 00 00 00
Dynamic Host Configuration Protocol: Protocol
Packets: 20281 · Displayed: 24 (0.1%)
Profile: Default

Capture 100 packets:-

1. Open wireshark.
2. Select the network interface.
3. Set the capture limit 100 packets.
4. Start capturing.
5. Stop and raise the captured packets.

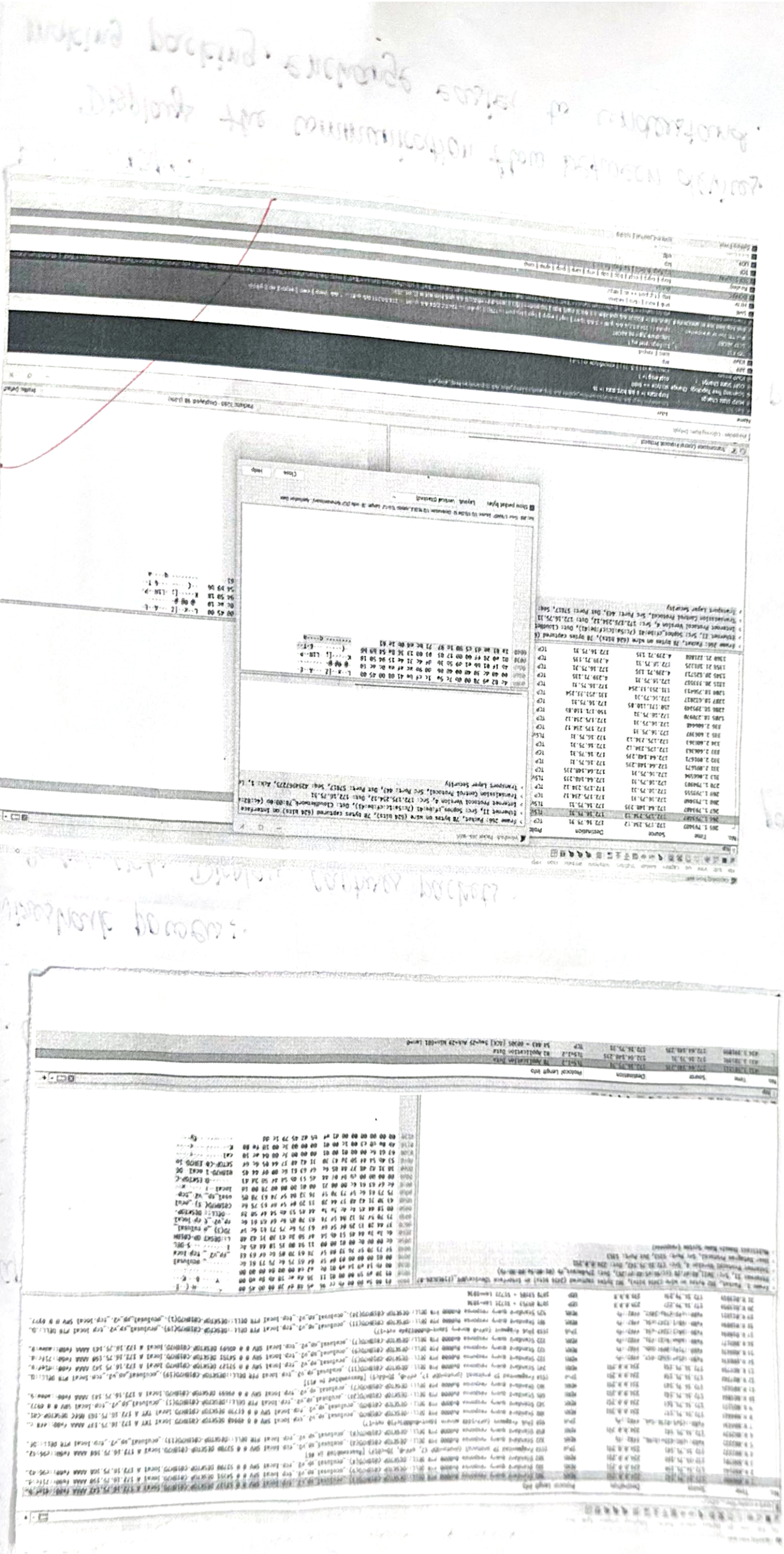
Analyzing Packets using wireshark tool:-

1. Create a filter to displays only TCP/UDP packets.
   * Open the capture file.
   * Apply the TCP || UDP filter.
   * Inspect and view the flow graph.

2. Create a file to display only ARP packets.
   * open and capture file.
   * Apply the arp filter.
   * Inspect and view the flow graph.

3. Create a filter to display only DNS packets.
   * Open the capture file.
   * Apply the DNS filter.
   * Inspect and view the flow graph.

4. Create a filter to display only IP/ICMP packets.
   * Open the capture file.
   * Apply the IP or ICMP filter.
   * Inspect and view the flow graph. and same capture.



5. Create a filter to display only DHCP packet.

* Open the capture file.
* Apply the DHCP file.
* Inspect and view the flow graph and save the capture.

RESULT:

Thus the feature of wireshark and the implement of different filter was successfully compiled.

dns

No.	Time	Source	Destination	Protocol	Lengtl	Info
5047	64.823837	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xf2bf HTTPS dns.google
5048	64.823968	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x2c40 A dns.google
5092	64.872708	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x2c40 A dns.goo
5093	64.872708	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xf2bf HTTPS dns
5907	73.507194	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xdd84 HTTPS dns.google
5908	73.507610	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x7b95 A dns.google
5909	73.522819	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x7b95 A dns.goo
5910	73.522819	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xdd84 HTTPS dns
6172	75.347431	172.16.75.31	8.8.8.8	DNS	70	Standard query 0x7402 HTTPS dns.google
6173	75.347571	172.16.75.31	8.8.8.8	DNS	70	Standard query 0xddfb A dns.google
6174	75.377779	8.8.8.8	172.16.75.31	DNS	102	Standard query response 0xddfb A dns.goo
6180	75.398557	8.8.8.8	172.16.75.31	DNS	146	Standard query response 0x7402 HTTPS dns

```
> Frame 5047: Packet, 70 bytes on wire (560 bits), 70 bytes
> Ethernet II, Src: CloudNetwork_78:00:0b (4c:82:a9:78:00:0
> Internet Protocol Version 4, Src: 172.16.75.31, Dst: 172.
> User Datagram Protocol, Src Port: 53918, Dst Port: 53
> Domain Name System (query)
```

```
0000  7c 5a 1c cf be 41 4c 82  a9 78 00 0b 08 00 45 00   |z
0010  00 38 65 85 00 00 80 11  e9 ee ac 10 4b 1f ac 10   ·8
0020  48 01 d2 9e 00 35 00 24  99 73 f2 bf 01 00 00 01   H
0030  00 00 00 00 00 00 03 64  6e 73 06 67 6f 6f 67 6c   ·
0040  65 00 00 41 00 01                                  e
```

arp

No.	Time	Source	Destination	Protocol	Lengtl	Info
1343	10.957901	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.75.125? Tell 172.16.75.6
1977	14.358615	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2098	15.471882	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2400	17.952584	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.78.5? Tell 172.16.75.61
2747	24.883561	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.75.204? Tell 172.16.75
4906	61.332363	CloudNetwork_77:ff:...	Broadcast	ARP	42	Who has 172.16.77.93? Tell 172.16.

Cisco Packet Tracer Student

File Edit Options View Tools Extensions Help

Logical [Root] New Cluster Move Object Set Tiled Background Viewport

Time: 00:00:06 Power Cycle Devices Fast Forward Time

Realtime

Routers

1841 1941 2620XM 2621XM

2620XM

Scenario 0

New Delete

Toggle PDU List Window

Fire Last Status Source Destination